Diploma in Software Testing and Automation

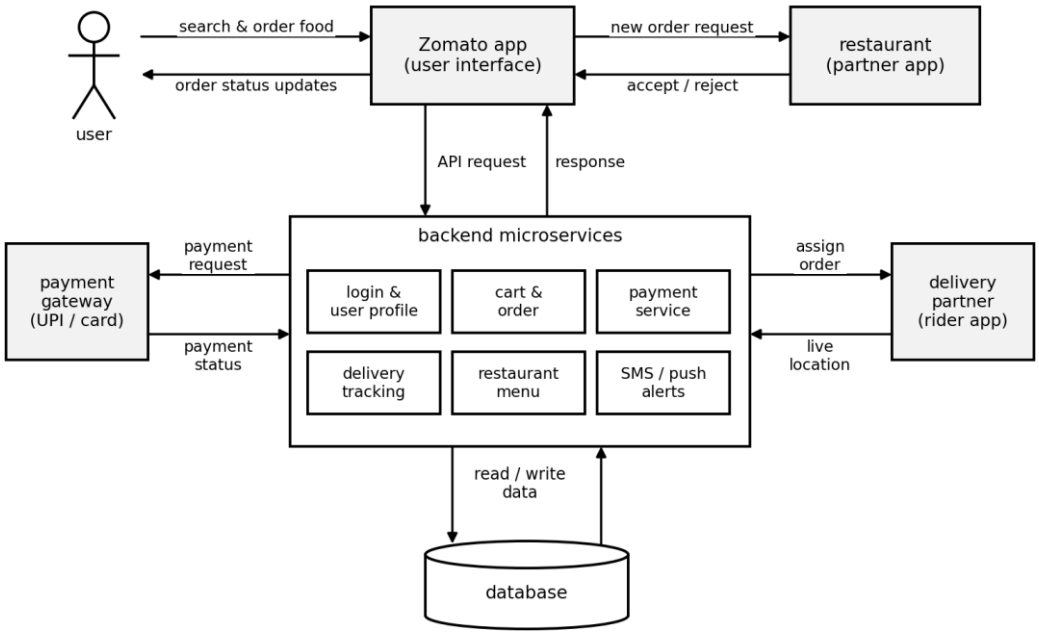
SESSION 1 - Introduction to Testing & Software Engineering

Module Assignment - Tasks 1 to 4

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Task 1: Food Delivery App Architecture Diagram

Draw a simple diagram showing the main components of a food delivery app like Zomato or Swiggy and label how they interact.



How the components interact:

- **User:** searches food in the Zomato app, places the order and gets order status updates.
- **Zomato app (user interface):** the screen the user sees. It sends the new order request to the restaurant and the API requests to the backend, and shows the responses back to the user.
- **Restaurant (partner app):** receives the new order and accepts or rejects it.
- **Backend microservices:** the main part, divided into small services: login & user profile, cart & order, payment service, delivery tracking, restaurant menu and SMS / push alerts. Each service does its own job.
- **Payment gateway (UPI / card):** gets the payment request from the payment service and sends back the payment status.
- **Delivery partner (rider app):** gets the assigned order and shares the live location while delivering.
- **Database:** the backend reads and writes user details, menus and orders here.

Task 2: Testing Activities Before Release (App: Paytm)

Pick any app you use daily and list 5 different testing activities that must be performed before the app is released to users.

I have chosen Paytm because I use it almost every day for payments and recharge. Before Paytm is released, these 5 testing activities should be done:

No.	Testing Activity	What is checked
1	Login / Authentication Testing	Login with mobile number and OTP works properly, wrong OTP is rejected, PIN / fingerprint lock works, and the session logs out after some time.
2	Payment Testing	UPI, wallet and bank transfers go through correctly. Checks for failed payments, money deducted but not received, refund, and wrong or duplicate amounts.
3	Notification Testing	SMS and push notifications for payment success, failure and cashback arrive on time and show the correct amount and details.
4	Performance / Speed Testing	The app should stay fast and not crash when lakhs of people pay at the same time, for example during festival offers or IPL days.
5	Security Testing	User data, card details and bank information must be encrypted. Checks that nobody can hack the account or misuse the data.

Task 3: Importance of Testing at Each Stage of SDLC (Example: IRCTC)

Explain in your own words why testing is important at each stage of the Software Development Life Cycle (SDLC), using an example from an app like BookMyShow or IRCTC.

Testing is not only done at the end. If we test at every stage of SDLC, mistakes are found early and fixing them costs much less. I am explaining this with the IRCTC train ticket booking app.

SDLC Stage	Why testing is important (IRCTC example)
1. Requirement Analysis	We review the requirements to see whether they are clear. For example, if the requirement just says 'book a ticket' but does not say how many passengers are allowed or what happens in Tatkal time, testers can raise this question early.
2. Design	The design is checked for problems. For example, will the system handle thousands of users booking at 10 AM for Tatkal? Reviewing the design here avoids big changes later.
3. Development (Coding)	Developers do unit testing on their own code, like checking that the seat availability function gives the right count, so small bugs are fixed before they spread.
4. Testing	Full testing is done: search trains, select class, fill passenger details, pay and get the PNR. Testers check that the ticket is not booked twice for the same seat and that money is refunded when booking fails.
5. Deployment	After going live, a quick check is done to make sure everything works on the real server, like login, payment and ticket download.

SDLC Stage	Why testing is important (IRCTC example)
6. Maintenance	When a new feature or update is added, regression testing checks that old things (like cancellation and refund) still work. Bugs reported by users are also tested after they are fixed.

Conclusion: Testing at every stage saves time and money, avoids failures like the app crashing during Tatkal booking, and gives users a smooth experience.

Task 4: Objectives and Purpose of Testing (New Music Streaming Feature in Spotify)

Write down the main objectives and purpose of testing by imagining you are part of a team launching a new music streaming feature in Spotify. List at least 3 objectives and explain each in one line.

Suppose our team is launching a new feature in Spotify, such as 'Collaborative Playlist with live listening'. The main objectives of testing are:

- **1. Find and fix bugs early:** to catch problems like songs not playing, wrong song order or app crash before users face them.
- **2. Check that the feature works as required:** to confirm that play, pause, skip, add song and share playlist work exactly as planned in the requirements.
- **3. Ensure good performance:** to make sure songs load quickly and stream without buffering, even on slow internet and when many users listen together.
- **4. Ensure security and privacy:** to protect user accounts, playlists and payment details from being leaked or misused.
- **5. Give a good user experience:** to confirm the feature is easy to use and works well on different phones, screen sizes and Android / iOS versions.

Purpose of testing: The main purpose is to deliver a good quality product, reduce the risk of failure after launch and build user trust, so that people enjoy the new feature without problems.